

BYTES CAFFE - STANDARD OPERATING PROCEDURE

gemini

Objective: Zero-leakage inventory and high-speed service.

Recipe Control: Every drink has a digital "Bill of Materials" (BOM). Selling one "Cappuccino" must automatically deduct 18g of coffee beans and 200ml of milk in the database.

Wastage Log: All milk/pastry expiration must be logged via a "Waste Entry" button before EOD to adjust profit margins accurately.

Opening/Closing: Baristas must complete a digital hygiene checklist (Machine backflush, Grinder calibration) before the POS unlocks for sales.

Chat gpt

Business Overview

Bytes Caffe operates as a modern coffee shop providing beverages, snacks, dine-in services, and customer engagement experiences.

Objectives

Deliver premium customer experience.

Maintain food quality and hygiene.

Ensure efficient service operations.

Increase customer retention.

Organizational Structure

Key Roles

Cafe Manager

Barista

Kitchen Staff

Cashier

Service Staff

Housekeeping

Procurement Staff

Opening Procedures

Daily Startup Checklist

Unlock premises.

Turn on POS and billing systems.

Prepare coffee machines.

Verify ingredient availability.

Check cleanliness.

Display menu boards.

Customer Service SOP

Greeting Standards

Welcome every customer within 10 seconds.

Assist with menu recommendations.

Maintain polite communication.

Order Taking

Confirm order clearly.

Verify customization requests.

Generate bill before preparation.

Beverage Preparation SOP

Coffee Standards

Follow standard recipes.

Use fresh ingredients.

Maintain temperature standards.

Ensure presentation quality.

Hygiene Requirements

Wash hands frequently.

Clean equipment regularly.

Store milk and perishables correctly.

Kitchen SOP

Food Preparation

Follow approved recipes.

Maintain hygiene standards.

Monitor cooking times.

Prevent cross-contamination.

Storage Standards

Label all items.

Follow FIFO method.

Maintain refrigeration logs.

Inventory SOP

Daily Inventory

Record opening stock.

Monitor ingredient usage.

Report shortages.

Minimize wastage.

Procurement

Purchase from approved vendors.

Verify invoice accuracy.

Inspect delivered goods.

Billing & Cash Handling SOP

POS Operations

Every order must be billed.

No manual billing without approval.

Provide customer receipts.

Cash Management

Shift-wise cash reconciliation.

Report discrepancies immediately.

Secure cash drawer.

Housekeeping SOP

Cleaning Schedule

Tables cleaned after every use.
Washrooms cleaned regularly.
Kitchen sanitized daily.
Waste disposed properly.

Customer Complaint SOP

Complaint Resolution

Listen actively.

Apologize professionally.

Replace defective products if required.

Escalate serious issues.

Closing Procedures

End-of-Day Activities

Clean equipment.

Shutdown machines.

Count cash.

Secure inventory.

Lock premises.

Claude thinking

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1. BUSINESS OVERVIEW

1.1 About Bytes Caffè

Bytes Caffè is the Food & Beverage arm of AEC Group — a modern, welcoming coffee shop serving quality coffee, beverages, snacks, and light meals. The café serves as a social hub for students, professionals, cinema-goers, and the local community.

1.2 Vision

To be the neighborhood's favorite café — known for great coffee, fresh food, and a vibrant atmosphere.

1.3 Mission

Serve consistently high-quality coffee and food using fresh, quality ingredients.

Create a comfortable, inviting space for work, study, and socializing.

Maintain the highest standards of hygiene and food safety.

Build a loyal customer base through exceptional service and community engagement.

1.4 Facility Layout

Area

Description

Dining Area

Indoor seating with tables, chairs, and lounge seating

Counter / Billing Area

Customer-facing order counter with display, POS

Coffee Bar

Espresso machine, grinder, brewing station

Kitchen / Prep Area

Food preparation, cooking, plating

Cold Storage

Refrigerators, freezer for perishables

Dry Storage

Packaged goods, dry ingredients, disposables

Restroom

Customer restroom facility

Staff Area

Employee lockers, break area

Wash Area

Dishwashing and sanitization station

1.5 Operating Hours

Day

Hours

Monday – Saturday

8:00 AM – 9:00 PM

Sunday / Holidays

9:00 AM – 9:00 PM

NOTE

Hours may be extended during special events, movie premieres at AEC Cinemas, or festive seasons with manager approval.

2. ORGANIZATION STRUCTURE & STAFFING

2.1 Reporting Structure

Caffe Manager

└─ Shift Supervisor / Senior Barista

- | | — Baristas (2)
- | | — Kitchen Staff / Cook (1-2)
- | | — Service Staff / Waiters (1-2)
- | | — Utility / Housekeeping (1)

2.2 Roles & Responsibilities

Role

Key Responsibilities

Caffe Manager

Overall operations, menu planning, P&L, vendor management, staff training, quality control

Shift Supervisor

Shift management, staff coordination, cash handling, opening/closing duties

Baristas

Coffee and beverage preparation, equipment maintenance, customer interaction

Kitchen Staff / Cook

Food preparation per recipe cards, kitchen hygiene, stock usage tracking

Service Staff

Table service, order taking, bill presentation, table clearing and resetting

Utility Staff

Dishwashing, cleaning, restocking, waste disposal

2.3 Shift Structure

Shift

Timing

Min Staff

Morning / Shift 1

7:30 AM – 3:00 PM

3 (1 barista, 1 cook, 1 service)

Evening / Shift 2

2:30 PM – 9:30 PM

3 (1 barista, 1 cook, 1 service)

Full Day (Manager)

9:00 AM – 6:00 PM (rotational off)

1

2.4 Dress Code

Caffe-branded apron (provided by management) worn at all times during shift.

Cap/hair net mandatory for kitchen and barista staff.

Clean, pressed clothing underneath; no open-toed shoes.

Name badge worn visibly.

No strong perfume, dangling jewelry, or long nails (food safety requirement).

3. DAILY OPERATIONS — OPENING

3.1 Pre-Opening Checklist (Shift Supervisor — 30 min before opening)

Facility & Cleanliness:

- Unlock premises; disarm alarm

- Turn on lights, AC/fans, background music system

- Inspect dining area: Tables wiped, chairs arranged, floor clean

- Restroom checked: Clean, stocked (soap, tissue, sanitizer)

- Outdoor seating (if any) arranged and cleaned

Kitchen & Prep:

- Inspect kitchen: Surfaces clean, no overnight pests/issues

- Check refrigerator temperatures (must be $\leq 4^{\circ}\text{C}$; freezer $\leq -18^{\circ}\text{C}$) — log in temperature sheet

- Verify expiry dates on all perishable stock

- Start mise en place for the day's menu:

- Vegetables washed and prepped

- Sauces and condiments portioned
- Batter/dough prepared (if applicable)
- Review daily specials with cook

Coffee Bar:

- Power on espresso machine (needs 15-20 min warm-up)

- Grinder calibrated: Pull a test shot (aim for 25-30 seconds extraction)

- Milk steaming wand purged and cleaned

- Coffee beans hopper filled; backup beans accessible

- All syrup bottles, toppings, and cups stocked at station

Billing & Administration:

- POS system powered on and tested (print a test receipt)

- Cash float counted and logged (amount as per policy)

- Card/UPI payment terminal tested

- Day's promotional offers entered into POS (if any)

Staff Briefing (5 minutes):

- Review today's specials, promotions, and any 86'd items (unavailable)

- Assign roles/stations for the shift

- Communicate any special events or expected large orders

4. FOOD PREPARATION & KITCHEN OPERATIONS

4.1 Recipe Standards

All items prepared strictly per Standard Recipe Cards (SRC). Each SRC contains: Ingredients with exact quantities, preparation method, portion size, plating guide, and photo reference.

Recipe deviations require Caffe Manager approval.

New items undergo a tasting and costing session before being added to the menu.

4.2 Portion Control

Portioning tools (scoops, ladles, measuring cups) used for consistency.

Pre-portioned items stored in labeled containers.

No "extra" portions given without manager approval (charged as applicable).

4.3 Menu Categories

Category

Examples

Hot Beverages

Espresso, Cappuccino, Latte, Americano, Hot Chocolate, Tea

Cold Beverages

Iced Coffee, Cold Brew, Smoothies, Milkshakes, Fresh Juice

Snacks

Sandwiches, Wraps, Toast, Cookies, Muffins, Croissants

Light Meals

Pasta, Salads, Rice Bowls, Burgers

Desserts

Cakes, Brownies, Waffles, Ice Cream

Specials

Daily special / Seasonal items

4.4 Food Temperature Standards

Type

Required Temperature

Hot food (serving)

$\geq 63^{\circ}\text{C}$

Cold food (serving)

$\leq 8^{\circ}\text{C}$

Refrigerator storage

$\leq 4^{\circ}\text{C}$

Freezer storage

≤ -18°C

Reheating

Must reach ≥ 75°C core temperature

IMPORTANT

Food that has been in the danger zone (8°C–63°C) for more than 2 hours must be discarded immediately.

4.5 Allergen Management

Allergen information available for all menu items.

Staff trained to ask: "Do you have any food allergies we should be aware of?"

Common allergens flagged on the menu: Nuts, Dairy, Gluten, Eggs, Soy.

Separate utensils used for allergen-free preparation when requested.

Any allergen incident reported immediately to Caffe Manager and documented.

5. BEVERAGE OPERATIONS

5.1 Coffee Standards

Beans: Use approved supplier's beans only; store in airtight containers away from light/heat.

Grind: Adjusted daily based on humidity and bean age. Test shot pulled at start of each shift.

Espresso: Target 18-20g dose, 25-30 second extraction, 36-40ml output.

Milk: Steamed to 60-65°C for lattes/cappuccinos (never exceed 70°C — scalds milk).

Latte Art: Encouraged for all milk-based drinks; staff trained regularly.

5.2 Coffee Machine Maintenance

Task

Frequency

Purge steam wand

After every use
Backflush group heads
Every 2 hours
Clean drip tray
Every 2 hours
Full machine cleaning (tablets + descale)
End of day
Professional servicing
Monthly (per AMC)
Grinder burr replacement
Per manufacturer schedule

5.3 Other Beverages

Tea: Freshly brewed to order; steep times per tea type (black: 3-5 min, green: 2-3 min).

Smoothies/Shakes: Made fresh to order; blender washed between orders.

Fresh Juice: Fruits washed thoroughly; juice served immediately, not stored.

Cold Brew: Prepared in advance (12-18 hour cold steep); labeled with prep date; discard after 5 days.

6. SERVICE STANDARDS

6.1 Customer Journey

Step

Standard

Entry

Greet warmly within 10 seconds: "Welcome to Bytes Caffè!"

Seating

Guide to available table or allow self-seating

Menu

Present menu within 1 minute; suggest specials/popular items

Order

Take order within 3 minutes of seating; confirm order back to customer

Serving

Beverages: $\leq$ 5 minutes. Snacks: $\leq$ 10 minutes. Meals: $\leq$ 15 minutes

Check-In

Visit table once after serving: "Is everything okay? Can I get you anything else?"

Bill

Present within 2 minutes of request; process payment within 3 minutes

Table Reset

Clear and reset table within 5 minutes of departure

Farewell

"Thank you for visiting Bytes Caffe! See you soon!"

6.2 Counter Service (Takeaway / Quick Service)

Greet customer at counter.

Take order and suggest add-ons ("Would you like a cookie with your coffee?").

Process payment.

Call customer by name when order is ready.

Hand over with a smile: "Enjoy your coffee!"

6.3 Dine-In Etiquette

Background music at comfortable volume (conversational level).

Tables wiped and reset immediately after guests leave.

Condiments (sugar, sweetener, napkins) replenished continuously.

Wi-Fi password displayed at each table / counter.

Charging points available where possible.

7. ORDER MANAGEMENT & POS

7.1 Order Flow

Order taken at counter or table.

Order entered into POS system with table number (dine-in) or "Takeaway" tag.

Kitchen Display System (KDS) or printed KOT (Kitchen Order Ticket) sent to kitchen/bar.

Barista/cook prepares order.

Order placed on the service counter or delivered to table.

POS updated when served.

7.2 Order Modifications

Modifications (customization, allergen requests) clearly noted on KOT.

Any item removed after preparation: Logged as waste with reason.

Complimentary items (replacement for quality issue) require supervisor authorization.

7.3 Delivery / Takeaway Orders (If Applicable)

Takeaway orders packed in branded packaging.

Delivery orders (via aggregator apps or direct) confirmed within 2 minutes.

Packaging ensures food temperature maintained during transit.

Delivery partner pickup within 15 minutes of order preparation.

8. HYGIENE & FOOD SAFETY

8.1 Personal Hygiene Standards

Handwashing mandatory: Before starting work, after handling raw food, after using restroom, after touching face/hair, after handling waste.

Hands washed for minimum 20 seconds with soap and warm water.

Hand sanitizer available at each station.

No eating, drinking, or chewing gum in kitchen/prep area.

Cuts or wounds must be covered with a blue waterproof bandage.

Staff feeling unwell (fever, vomiting, diarrhea) must not work with food — inform manager immediately.

8.2 Kitchen Hygiene

Task

Frequency

Work surface sanitization

Every 2 hours + after raw food contact

Cutting boards cleaned

After each use; separate boards for veg/non-veg

Floor mopping (kitchen)

Every 3 hours + at close

Equipment cleaning (grills, fryers)

After each use + deep clean at close

Dishwashing sanitization

Continuous

Refrigerator cleaning

Weekly

Deep kitchen clean

Weekly (Saturday night or designated day)

8.3 Pest Control

Monthly pest control treatment by licensed provider.

No food left uncovered overnight.

Waste bins emptied before closing; no overnight accumulation.

Door screens/curtains maintained to prevent pest entry.

Any pest sighting reported immediately; area closed until treated.

8.4 Food Safety Compliance

FSSAI license displayed prominently at the premises.

All staff hold valid Food Handler's Certificate.

Temperature logs maintained 3 times daily (morning, afternoon, closing).

Health inspection readiness maintained at all times.

Annual food safety audit by external agency.

9. INVENTORY & SUPPLY CHAIN MANAGEMENT

9.1 Stock Management

Task

Frequency

Responsible

Opening stock count

Daily

Shift Supervisor

Closing stock count

Daily

Shift Supervisor

Stock audit & reconciliation

Weekly

Caffe Manager

Full inventory count

Monthly

Caffe Manager + Finance

9.2 Ordering Process

Shift Supervisor reviews stock levels against par stock levels (minimum required stock).

Items below par flagged on the Reorder Sheet.

Caffe Manager approves and places orders with approved vendors.

Deliveries accepted only with matching Purchase Order (PO).

Delivery checked: Quantity, quality, expiry dates, temperature (for cold items).

Stock entered into inventory system; GRN (Goods Received Note) issued.

Items stored per FIFO principle; new stock placed behind existing stock.

9.3 Supplier Management

Approved vendor list maintained by Caffe Manager.

Minimum 2 approved vendors per critical category (coffee beans, dairy, produce).

Vendor evaluation conducted quarterly (quality, delivery reliability, pricing).

Payment terms: As per negotiated agreement; processed via AEC Group Finance.

9.4 Wastage Management

Daily wastage log maintained: Item, quantity, reason (expired/damaged/prepared-not-sold/spillage), time.

Wastage reviewed weekly by Caffe Manager.

Target: Total wastage $\leq$ 5% of daily preparation cost.

High-wastage items reviewed for demand accuracy; menu or prep adjustments made.

10. DAILY OPERATIONS — CLOSING

10.1 Closing Checklist (Shift Supervisor)

Kitchen & Food:

- All leftover prepared food disposed of or stored per food safety rules

- All raw ingredients stored correctly (labeled with date)

- Refrigerator and freezer temperatures checked and logged

- Kitchen surfaces, equipment, and floors deep-cleaned

- Gas supply turned off (if applicable)

- Exhaust fan cleaned / filters checked

Coffee Bar:

Espresso machine fully cleaned (backflush, wipe, drain)

Grinder cleaned; hopper emptied if overnight

All milk jugs, cups, and equipment washed and dried

Coffee bar surfaces wiped and sanitized

Dining Area:

All tables cleared, wiped, and reset

Floor swept and mopped

Condiment stations restocked for next day

Chairs pushed in; furniture arrangement tidy

Billing & Cash:

POS closed; Daily Sales Report generated and printed

Cash counted and reconciled with POS total

Discrepancies (if any) noted and reported to Caffe

Manager

Cash secured in safe

Credit/UPI settlements verified

Facility:

Restroom cleaned and stocked

All waste disposed of in designated outdoor bins

AC/fans off; lights off (except security lights)

Doors and windows locked

CCTV verified recording

Alarm system armed

Reporting:

Daily Closing Report sent to Caffe Manager with:
Sales, cash reconciliation, wastage, stock notes,
issues/incidents

11. CUSTOMER EXPERIENCE & FEEDBACK

11.1 Complaint Handling

Listen actively without interrupting.

Apologize sincerely: "I'm sorry about that. Let me fix this for you."

Resolve immediately:

Quality issue → Replace the item (no charge).

Service delay → Offer a complimentary coffee/cookie.

Billing error → Correct and re-issue bill.

Escalate to Supervisor/Manager if unresolved.

Document in the Feedback Register: Date, customer name (if given), issue, action taken.

11.2 Feedback Collection

Feedback cards available at tables.

Google Review QR code displayed at billing counter.

Monthly feedback analysis by Caffe Manager — trends identified and action items created.

Top 3 improvement areas addressed each month.

11.3 Loyalty Program (If Implemented)

Stamp card: Buy 9 coffees, get the 10th free.

Tracked digitally or via physical stamp card.

Points system (if digital): ₹1 spent = 1 point; 100 points = ₹10 discount.

12. FINANCIAL OPERATIONS

12.1 Revenue Streams

Dine-In Sales — Primary revenue

Takeaway Sales — Growing segment

Delivery Sales — Via aggregator apps (Swiggy/Zomato) or direct

Catering / Bulk Orders — For AEC Group events and external orders

Cross-Promotion Revenue — Cinema combo tie-ups, Residency guest vouchers

12.2 Pricing & Costing

Target food cost: $\leq 30\%$ of selling price.

Target beverage cost: $\leq 25\%$ of selling price.

Menu pricing reviewed quarterly.

Cost analysis done for each new menu item before launch.

All pricing changes approved by Caffe Manager + Group Finance.

12.3 Reporting

Report

Frequency

Submitted To

Daily Sales & Cash Report

Daily

Caffe Manager

Weekly Revenue & Wastage Summary

Weekly

Caffe Manager + Group Finance

Monthly P&L Statement

Monthly

Group Finance + Group Director

Inventory Valuation Report

Monthly

Group Finance

Vendor Payment Summary

Monthly

Group Finance

13. MARKETING & PROMOTIONS

13.1 Regular Promotions

Happy Hour: 20% off beverages during 3-5 PM (weekdays).

Combo Deals: Coffee + Snack at bundled price.

Student Special: 10% off with valid student ID.

Weekend Brunch Menu: Special menu items on Sat-Sun mornings.

Seasonal Specials: Festival-themed drinks/snacks (e.g., Monsoon Coffee, Christmas Cookies).

13.2 Social Media

Daily post with café ambiance, featured drink, or food photo.

Instagram stories of preparation / behind-the-scenes.

Customer features (with permission) and reviews shared.

All posts approved by AEC Group Marketing.

13.3 Cross-Promotions with AEC Group

AEC Cinemas: Movie ticket holders get 10% off at Bytes Caffé.

AEC Residency: Welcome beverage voucher for hotel guests.

AEC Institute / Studies: Student discount; café as a study-friendly space during off-peak hours.

14. FACILITY MAINTENANCE & HOUSEKEEPING

14.1 Cleaning Schedule

Area

Frequency

Dining tables & chairs

After each customer + every hour

Floors (dining area)

Every 3 hours + end of day

Counter & display area

Every 2 hours

Kitchen floors

Every 3 hours + deep clean at close

Restroom

Every 3 hours

Windows & glass doors

Daily

Full deep clean

Weekly

14.2 Equipment Maintenance

Equipment

Maintenance

Frequency

Espresso machine

Professional service

Monthly

Grinder

Burr inspection + cleaning

Weekly

Refrigerator / Freezer

Temperature checks, coil cleaning

Daily checks; monthly coil clean

Cooking equipment

Inspection + deep clean

Weekly

AC units

Filter cleaning, servicing

Monthly filter; quarterly service

POS hardware

Software updates, cleaning

Monthly

15. SAFETY & EMERGENCY PROCEDURES

15.1 Fire Safety

Fire extinguisher locations: Kitchen (1), Counter area (1), Dining area (1).

Fire blanket in kitchen near cooking station.

Staff trained in extinguisher use.

Evacuation route posted; drill conducted quarterly.

15.2 Fire Emergency

Turn off gas and electrical cooking equipment immediately.

Use fire extinguisher/blanket for small fires.

If fire is uncontrollable: Evacuate all customers and staff via exits.

Call fire services (101 / 112).

Assembly at designated point; headcount by Supervisor.

15.3 Medical Emergency

First aid kit at the counter/manager's area.

At least 1 first-aid trained staff per shift.

Call 112 for serious incidents.

Incident report filed within 2 hours.

15.4 Gas Leak

Do not operate any switches (electrical sparks risk).

Turn off gas supply at mains immediately.

Open all windows and doors for ventilation.

Evacuate premises.

Call gas supplier / emergency services.

No re-entry until cleared by professionals.

15.5 Slip / Fall Incidents

Wet floor signs placed immediately after mopping.

Spills cleaned within 2 minutes.

Anti-slip mats in kitchen and wash areas.

Any customer/staff injury documented in Incident Register.

16. HR & STAFF POLICIES

16.1 Attendance & Punctuality

Biometric login/logout for all staff.

Arrive 15 minutes before shift for prep and briefing.

3 late arrivals in a month = written warning.

Leave requests: 48 hours advance notice (except emergencies).

16.2 Training

Training

Duration

Frequency

Induction (new staff)

3 days

At joining

Barista skills (espresso, latte art)

2 days

At joining + quarterly refresher

Food safety & hygiene

1 day

At joining + annual refresher

Customer service

Half day

Quarterly

Fire safety & emergency

Half day

Quarterly

New menu item training

As needed

Per menu update

16.3 Performance Review

Monthly review by Caffè Manager.

Metrics: Attendance, customer feedback, speed of service, upselling, hygiene compliance.

Employee of the Month recognition.

Underperformance handled per AEC Group HR progressive discipline policy.

16.4 Staff Meals & Benefits

Staff entitled to 1 meal and 2 beverages per shift (from designated staff menu).

No consumption of display/sale items without authorization.

Staff discount: 30% off on personal purchases (off-duty).

17. KEY PERFORMANCE INDICATORS

KPI

Target

Measurement

Daily Revenue

As per monthly target

Daily POS report

Food Cost %

$\leq 30\%$ of food revenue

Monthly P&L

Beverage Cost %

$\leq 25\%$ of beverage revenue

Monthly P&L

Customer Wait Time

Beverages ≤ 5 min; Food ≤ 15 min

Spot checks

Customer Satisfaction Score

$\geq 4.3 / 5$

Monthly feedback/reviews

Table Turnover Rate

$\geq 3x$ during peak hours

Observation

Wastage %

$\leq 5\%$ of daily prep cost

Weekly wastage report

Cash Discrepancy

$\leq 0.5\%$

Daily reconciliation

Staff Punctuality

$\geq 95\%$

Monthly biometric report

Hygiene Audit Score

$\geq 90\%$

Monthly internal audit
Google Review Rating
≥ 4.2 stars
Monthly check

PerplexityBYTES CAFFE - COMPREHENSIVE STANDARD OPERATING PROCEDURES MANUAL

Modern Coffee Shop with "Super Computer ERP" Integration
Version 2.0 | AEC Group | May 2026

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1. BUSINESS OVERVIEW

1.1 Facility Specifications

Area
Capacity
Equipment
Dining

45 seats (12 tables)
AC, WiFi, charging points
Counter
8m service counter
2 POS terminals, display case
Coffee Bar
Espresso station
2-group La Marzocco, grinder
Kitchen
25sqm
Convection oven, grill, fridge
Storage
Dry + Cold
IoT weight sensors, 4°C fridge
Hours: 8AM-9PM (Mon-Sat), 9AM-9PM (Sun/Holidays)
1.2 Revenue Model
Beverages: 60% (Target COGS $\leq$ 25%)
Food/Snacks: 35% (Target COGS $\leq$ 30%)
Combos: 5% uplift via bundling
F&B Attach Target: ₹185 Average Order Value (AOV)

2. ORGANIZATION & STAFFING

2.1 Shift Structure (IoT Biometric)

text

MORNING (7:30AM-3PM): 4 staff

- Supervisor (1), Barista (1), Kitchen (1), Service (1)

EVENING (2:30PM-9:30PM): 5 staff

- Supervisor (1), Barista (2), Kitchen (1), Utility (1)

MANAGER: 9AM-6PM (rotational)

2.2 Role Matrix

Role
ERP Access
Critical Tasks
Manager
Full
P&L, AI reports, vendor
Supervisor
Ops
POS unlock, inventory
Barista
POS
BOM auto-deduct, waste log
Kitchen
POS
Recipe compliance

3. DAILY OPERATIONS - OPENING (30 MIN PRE-OPEN)

3.1 Digital Checklist (ERP Tablet - POS Locked Until Complete)

text

CRITICAL (Photos Required):

- ☐ Espresso machine backflush complete [PHOTO]
- ☐ Grinder calibrated (25-30s test shot) [PHOTO]
- ☐ Milk fridge $\leq 4^{\circ}\text{C}$, beans hopper filled [PHOTO]
- ☐ POS cash float ₹3000 verified [PHOTO]
- ☐ Hygiene checklist 100% [PHOTO UPLOAD]

IoT AUTO-CHECK:

- Weight sensors online (beans/milk)
- Fridge temps logged ($\leq 4^{\circ}\text{C}$)
- CCTV + fire alarm active

3.2 Machine Pre-Heat Protocol

text

La Marzocco: 20 min warm-up → 93°C brew temp

Grinder: Zero retainers, dose 18g espresso

Milk: Steam wand purged x3

Oven: 180°C convection calibration

4. BEVERAGE PREPARATION (BOM Auto-Deduct)

4.1 Digital Bill of Materials (Every Sale Triggers)

text

CAPPUCCINO (₹120):

- Coffee beans: 18g ☒ [IoT deduct]
- Milk: 200ml ☒ [Weight sensor]
- Cup: 1x 200ml ☒ [POS count]
- Lid/Sugar: Auto-tracked

ICED LATTE (₹140):

- Espresso: 2 shots (36g)
- Milk: 150ml cold
- Ice: 120g
- Syrup: 15ml (flavor-specific)

4.2 Quality Standards

text

ESPRESSO: 25-30s extraction, 36-40ml, 93°C

MILK FOAM: 60-65°C microfoam, glossy surface

ICED DRINKS: ≤8°C serve temp

TEA: 3-5min steep (type-specific)

4.3 Machine Hygiene (Per 50 Orders)

text

- ☐ Steam wand purge + wipe [30s]
- ☐ Portafilter clean + dry

- ☐ Drip tray empty
- ☐ Group head backflush

5. FOOD PREPARATION

5.1 Recipe Compliance (Photo Verification)

text

SANDWICH (₹99):

1. Bread: 120g toasted
2. Filling: 100g exact (chicken/veggie)
3. Cheese: 20g slice
4. Sauce: 15ml measured
5. Wrap + cut → [PHOTO] → Serve

PASTRY DISPLAY:

- Expiry sticker visible
- FIFO rotation enforced
- Daily waste logged before 6PM

5.2 Temperature Control

text

HOT FOOD: Serve $\geq 63^{\circ}\text{C}$ (probe check)

COLD FOOD: Serve $\leq 8^{\circ}\text{C}$

FRIDGE: $\leq 4^{\circ}\text{C}$ (IoT logged 3x daily)

FREEZER: $\leq -18^{\circ}\text{C}$

REHEAT: $\geq 75^{\circ}\text{C}$ core temp

6. ORDER MANAGEMENT & POS

6.1 Zero-Cash Workflow (98% UPI)

text

1. Customer orders → POS touchscreen
2. BOM auto-shown → Confirm → Pay (Razorpay)
3. Kitchen Display System (KDS) lights up

4. "Order #23 Ready" → Customer name called
 5. Pickup photo → Table delivery optional
- ## 6.2 Dynamic Pricing (AI-Driven)

text

PEAK (4-7PM): +15% all items

LOW DEMAND (<30% tables): -10%

WEEKDAY 2-4PM: Coffee -20%

CROSS-PROMO: Cinema ticket scan = 10% off

7. INVENTORY & BOM TRACKING

7.1 IoT Zero-Leakage System

text

WEIGHT SENSORS (Real-time):

- Coffee beans: 25kg hopper → Auto-reorder <5kg
- Milk: 20L tetras → Alert <4L
- Flour/Sugar: Dry storage bins

DAILY RECONCILIATION:

POS Sales BOM vs Weight Sensor Delta

Discrepancy >2% → Manager investigation

7.2 Waste Logging (Mandatory EOD)

text

WASTE BUTTON (POS):

- Milk expiry: Scan tetra → 2L logged
- Pastry stale: Photo + weight → 500g logged
- Spillage: Photo evidence required

Target: <3% daily COGS

8. CUSTOMER SERVICE

8.1 45-Second Service Cycle

text

1. GREET: "Hi! Coffee or our special today?" [10s]
2. ORDER: Confirm + upsell [20s]
3. PAY: UPI QR scan [10s]
4. PREP: KDS + call name [5s]

Target AOV: ₹185+

8.2 Loyalty Program

text

QR SCAN (Repeat customers):

- 5th coffee free
- Birthday month: Free pastry
- 10 orders: ₹50 voucher

9. DAILY OPERATIONS - CLOSING (30 MIN)

9.1 Mandatory ERP Steps

text

REQUIRED (Photos):

- ☐ La Marzocco full backflush [PHOTO]
- ☐ Grinder empty + clean [PHOTO]
- ☐ Fridge temp + expiry scan [PHOTO]
- ☐ Cash reconciliation video [PHOTO]
- ☐ Waste % logged <3% [PHOTO]

IoT AUTO:

- Weight sensors zeroed
- POS closed → P&L generated
- 6AM AI report queued

10. HYGIENE & FOOD SAFETY

10.1 FSSAI Compliance Matrix

Check

Frequency

Method

Hand wash

Every 30min

20s soap + warm water

Surface wipe

Post-raw food

Food-grade sanitizer

Gloves

Every 1h

Powder-free, single-use

Fridge temp

3x daily

IoT + manual log

11. FINANCIAL OPERATIONS

11.1 Auto P&L Generation

text

REVENUE: POS total (₹42k target/day)

COGS: BOM tracked ($\leq 28\%$ target)

LABOR: Biometric hours $\times$ wage

UTILITIES: IoT meter (₹2k target/day)

NET PROFIT: Real-time ERP dashboard

12. SAFETY & EMERGENCY

12.1 Gas Leak Protocol (30s)

text

1. GAS OFF at mains

2. NO SWITCHES (spark risk)

3. EVACUATE + call 112

4. Ventilate 30min minimum

13. AI SUPER COMPUTER INTEGRATION

13.1 Daily 6AM WhatsApp

text

"BYTES CAFFE YESTERDAY:

Revenue: ₹41,800 (AOV ₹187 ↑2%)

Inventory: Beans 23% used, Milk waste 2.1%

⚠️ ALERTS:

- Grinder calibration due
- Pastry waste 4.2% (review prep)

💡 IDEAS:

1. Launch ₹149 coffee+sandwich combo
2. Happy hour 4-6PM +15% tables
3. Cinema cross-promo QR codes"

14. KEY PERFORMANCE INDICATORS

KPI

Target

ERP Auto-Track

AOV

₹185+

Real-time

Inventory Accuracy

98%+

IoT vs POS

Waste %

<3%

Daily logged

Table Turnover

3x peak

Sensor count
UPI %
95%+
Razorpay data

BYTES CAFFE - KEY POINTS SUMMARY

Facility: 45-seat coffee shop, 8AM-9PM, modern espresso bar + kitchen. Staffing: 4-5/shift (Supervisor, Barista, Kitchen, Service). Revenue: Beverages 60%, Food 35%, Target AOV ₹185. Automation: IoT weight sensors, BOM auto-deduct, Razorpay UPI. Core SOPs: Digital hygiene checklists, waste logging, machine backflush. KPIs: Inventory 98% accuracy, waste <3%, table turnover 3x peak.

SUPER MASTER PROMPT: BYTES CAFFE ERP (AEC Group Module 2)

Build production-ready Bytes Caffe ERP using Django 5+ DRF, React 18+ Vite, Neon PostgreSQL (django-tenants ready for AEC Super App). Mobile-first PWA with glassmorphism UI, 98% automation via IoT/Razorpay/WhatsApp.

🔧 Tech Stack

text

BACKEND: Django+DRF+Celery[redis]+Channels

FRONTEND: React+Vite+Tailwind+Recharts+React-Query

DB: Neon PostgreSQL (tenant: bytes_caffe)

INTEGRATIONS: Razorpay, WhatsApp Business API, MQTT IoT

AI: Perplexity API (6AM reports)

DEPLOY: Docker-compose + Railway/Neon



24 Core Pages (Mobile-First)

Dashboards

Owner P&L: AOV ₹187 trends, waste %, inventory accuracy

Manager Ops: Live table occupancy, peak hour alerts

Staff Tasks: Personal POS/waste/cleaning checklists

POS & Orders

Mobile POS: 3-tap checkout (scan→pay→KDS)

Kitchen Display: Real-time order queue

Dynamic Pricing: Peak +15%, Happy Hour -20%

Inventory (IoT)

BOM Tracker: Cappuccino=18g beans+200ml milk auto-deduct

Weight Sensors: Beans/milk real-time, auto-reorder <20%

Waste Logging: Photo + weight mandatory EOD

Operations

Machine Health: Espresso backflush scheduler

Hygiene Checklists: Photo-verified, POS unlock blocker

Fridge Monitoring: $\leq 4^{\circ}\text{C}$ IoT alerts



Database Schema

python

```
class RecipeBOM(models.Model):
```

```
    name = CharField() # "Cappuccino"
```

```
    beans_grams = FloatField() # 18g
```

```
    milk_ml = FloatField() # 200ml
```

```
    price = DecimalField() # ₹120
```

```
class Inventory(models.Model):
```

```
    item = CharField() # "Coffee Beans"
```

```
iot_weight_kg = FloatField() # Real-time  
par_stock_kg = FloatField() # Auto-reorder
```

```
class Transaction(models.Model):  
    recipe = ForeignKey(RecipeBOM)  
    qty = IntegerField()  
    total = DecimalField()  
    table_no = CharField(blank=True)  
    waste = BooleanField(default=False)
```

🔑 Zero-Touch Integrations

Razorpay: /webhook/ → Transaction + BOM deduct

WhatsApp: Inbound leads, 6AM AI reports

MQTT

IoT: cafe/beans_weight, cafe/milk_tetra, cafe/fridge_temp

🧠 AI Engine (Daily)

python

6AM Celery: "AOV ₹187↑, waste 2.1%, grinder due"

Deliver COMPLETE mobile-responsive PWA
codebase. Glassmorphism UI, Kerala-localized, AEC Group
tenant-ready. Launch in 48hrs.